

Raguraja Krishnan Natarajan Mangaleshwaran

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EDUCATION

State University of New York - Binghamton

Binghamton, NY

Master of Science - Information Systems with Applied Data Science; GPA: **3.33/4.00**

Aug 2024 - Present

- **Coursework:** Python With Data Mining, Machine Learning, Tools for Data Science, Data Visualization Tools, Databases Backend, Cybersecurity

Vellore Institute of Technology

Bangalore, India

Post Graduate Program - Data Science; GPA: **3.80/4.00**

Aug 2023 - Jun 2024

- **Coursework:** Data Structures and Algorithms, Python for Data Analytics, Big Data, Deep Learning, DBMS, R, SQL

Sri Ramachandra Faculty of Engineering and Technology

Chennai, India

Bachelor of Science - Data Science; GPA: **3.06/4.00**

Aug 2020 - July 2023

- **Coursework:** Computer Networks, C Programming, Cloud Computing, Calculus, AI, Java, IoT, Mobile Programming, Design and Analysis of Algorithms

TECHNICAL SKILLS

Technical Areas: Data Analytics, Exploratory Data Analysis (EDA), Machine Learning, Deep Learning, Data Visualization, Statistical Modeling, Cloud Computing, Database Management Systems (DBMS), Data Preprocessing, Transfer Learning

Programming Languages: Python (NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow), SQL, R, HTML, CSS

Tools & Platforms: Jupyter Notebook, Google Colab, Tableau, Power BI, AWS (Foundational), Azure Custom Vision, Git, Excel

Soft Skills: Critical Thinking, Analytical Problem Solving, Team Collaboration, Effective Communication, Adaptability

EXPERIENCE

Data Analyst Intern

Chennai, India

Indian Orthocare Surgicals (Stryker Distributor)

Oct 2023 - Dec 2023

- Automated daily reporting dashboards using **Power BI**, reducing manual effort by 60%
- Integrated Microsoft Azure for **model deployment**, increasing defect detection accuracy by 27%.
- Streamlined quality control workflows, reducing manual inspection time by 30%.

Data Science Intern

Chennai, India

Nokia Solutions and Networks

May 2023 - Jul 2023

- Spearheaded a real-time **Video Analytics** project using **YOLO v3**, **computer vision**, and **machine learning**.
- **Integrated Microsoft Azure** for model deployment, increasing defect detection accuracy by **27%**.
- Streamlined **quality control workflows**, reducing manual inspection time by **30%**.

Data Analyst Intern

Chennai, India

Rankraze Technologies

Apr 2022 - Jun 2022

- Developed a custom **Python**-based SEO analysis tool inspired by **SEMrush**, performing **140+** **technical**, content, and performance checks.
- **Integrated Google PageSpeed API**, keyword density analysis, and robot.txt parser for detailed SEO reports.
- **Enhanced UX** by building an intuitive interface and exportable audit reports.

PROJECTS

Blood Group Detection Using Image Processing

Jan 2025 – Mar 2025

- **Built** an 8-class classifier using **CNN** to detect A, B, O, AB, A-, B-, O- and AB- groups from fingerprint images.
- **Implemented** hybrid **quantum-classical models** with **PennyLane + Qiskit**.
- **Achieved** strong generalization via fingerprint augmentation and **deep CNNs**.

Predictive Maintenance of Metro Train APUs using Machine Learning and PowerBI

Nov 2023 – Jan 2024

- Developed a **predictive maintenance framework** for metro train **APU systems** using sensor data from over **1.5 million time-stamped records**.
- Engineered features and applied **machine learning (Random Forest, Decision Tree)** and **deep learning models** to forecast failures with high accuracy.
- Leveraged **Power BI** for **interactive visual analytics** to identify key failure indicators and integrated **time series models** for early anomaly detection.

Airline Passenger Satisfaction Analysis in Tableau

Feb 2023 – April 2024

- **Cleaned** airline data using **Python** and **Excel**.
- **Created** interactive dashboards in **Tableau** for delay, satisfaction, and trends.
- **Applied clustering** to segment feedback and identify improvement areas.

CERTIFICATIONS

Free Code Camp: Data Analysis with **Python**

Free Code Camp: Machine Learning with **Python**

Google: Prompt Design in Vertex AI Skill Badge

AWS Academy Graduate: AWS Academy Introduction to Cloud Semester 1

Coursera: Business Analysis & Process Management

AWS Academy Graduate: AWS Academy Cloud Foundations